EEE533 Lab 1: Introductory Concepts

Saif Elsaady

EEE533: Semiconductor Device and Process Modeling

Problem 1: Crystal Lattice Nearest Neighbors (0.5 points)

Problem Statement:

What is the number of nearest-neighbor atoms for simple cubic, FCC and BCC lattice?

What is the number of the second nearest neighbors for simple cubic, FCC and BCC lattice?

Questions to answer:

• How many atoms directly touch each atom in simple cubic structure?

• How many atoms directly touch each atom in FCC structure?

• How many atoms directly touch each atom in BCC structure?

• What are the second nearest neighbors for each structure?

Results:

Simple Cubic: 6 nearest neighbors, 12 second nearest neighbors

FCC: 12 nearest neighbors, 6 second nearest neighbors

BCC: 8 nearest neighbors, 6 second nearest neighbors

Lattice Type	Nearest Neighbors	Second Nearest Neighbors
Simple Cubic	6	12
FCC	12	6
BCC	8	6

Problem 2: Primitive Cell Volumes (0.5 points)

Problem Statement:

Assume that a lattice constant a = 0.5 nm

Find the volume of a primitive cell for simple cubic, FCC and BCC crystal structures

Questions to answer:

• What is the relationship between lattice constant and primitive cell volume for simple cubic?

• What is the relationship between lattice constant and primitive cell volume for FCC?

• What is the relationship between lattice constant and primitive cell volume for BCC?

• Calculate the actual volumes given a = 0.5 nm

Results:

Lattice constant a = 0.5 nm

Simple Cubic primitive cell volume: 0.125 nm³

FCC primitive cell volume: 0.03125 nm³

BCC primitive cell volume: 0.0625 nm³

Lattice Type	Volume Formula	Volume (nm³)
Simple Cubic	a^3	0.125000
FCC	$a^3/4$	0.031250
BCC	$a^3/2$	0.062500

Problem 3: Temperature Dependence of Electron Density (2.0 points)

Problem Statement:

Consider bulk silicon material with donor doping ND=10^18 cm^-3

The donor ionization energy is ED = 50 meV

Plot the temperature dependence of the electron density

On the same axes, plot the temperature dependence of intrinsic carrier concentration

Include partial ionization (incomplete ionization) leading to carrier freeze-out at low temperatures

Use bisection method to solve the transcendental equation

Key Physics Concepts Implemented:

• Intrinsic carrier concentration

• Charge neutrality with partial ionization

• Fermi-Dirac statistics for donor ionization

• Transcendental equation solving using bisection method

• Temperature dependence from ~5K to 700K

$$n_i = \sqrt{N_c N_v} \exp\left(\frac{-E_g}{2k_B T}\right)$$

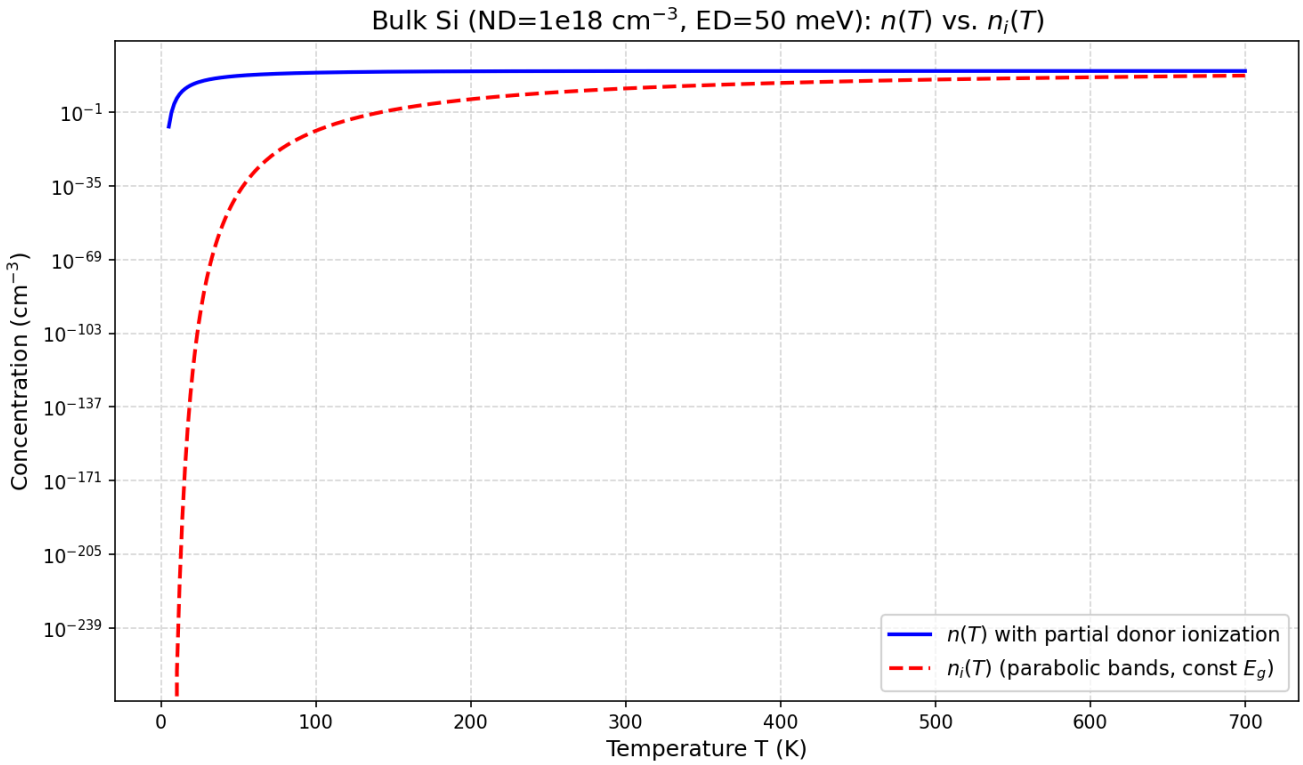
Charge Neutrality:

$$n = p + N_D^+$$

Donor Ionization (Fermi-Dirac):

$$N_D^+ = \frac{N_D}{1 + \frac{1}{g_D} \exp\left(\frac{E_F - E_D}{k_B T}\right)}$$

Results:
At 300 K: n ≈ 9.004e+17 cm^-3, ni ≈ 8.798e+09 cm^-3



Temperature dependence of electron density showing carrier freeze-out at low temperatures and approach to intrinsic behavior at high temperatures

Problem 4: Density of States in Different Dimensions (3.0 points)

Problem Statement:

(a) Derive gc,2D(E) in a quasi-2D system.

(b) Plot the DOS for 3D, 2D, and 1D systems.

Use ε(2D) = [50, 80, 95] meV and ε_i(1D) = [40, 75, 90, 95] meV.

Use energy in eV and length in cm.

Theoretical Background:

For parabolic bands, the conduction-band DOS in 2D is *constant* above each occupied subband edge and zero below it.

3D Density of States:

$$g_{c,3D}(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c}\cdot\theta(E - E_c)$$

2D Density of States per subband:

$$g_{c,2D}^{(i)}(E) = \frac{m^*}{\pi\hbar^2}\theta(E - E_c - \varepsilon_i)$$

Total 2D DOS (sum over subbands):

$$g_{c,2D}(E) = \sum_i \frac{m^*}{\pi\hbar^2}\theta(E - E_c - \varepsilon_i)$$

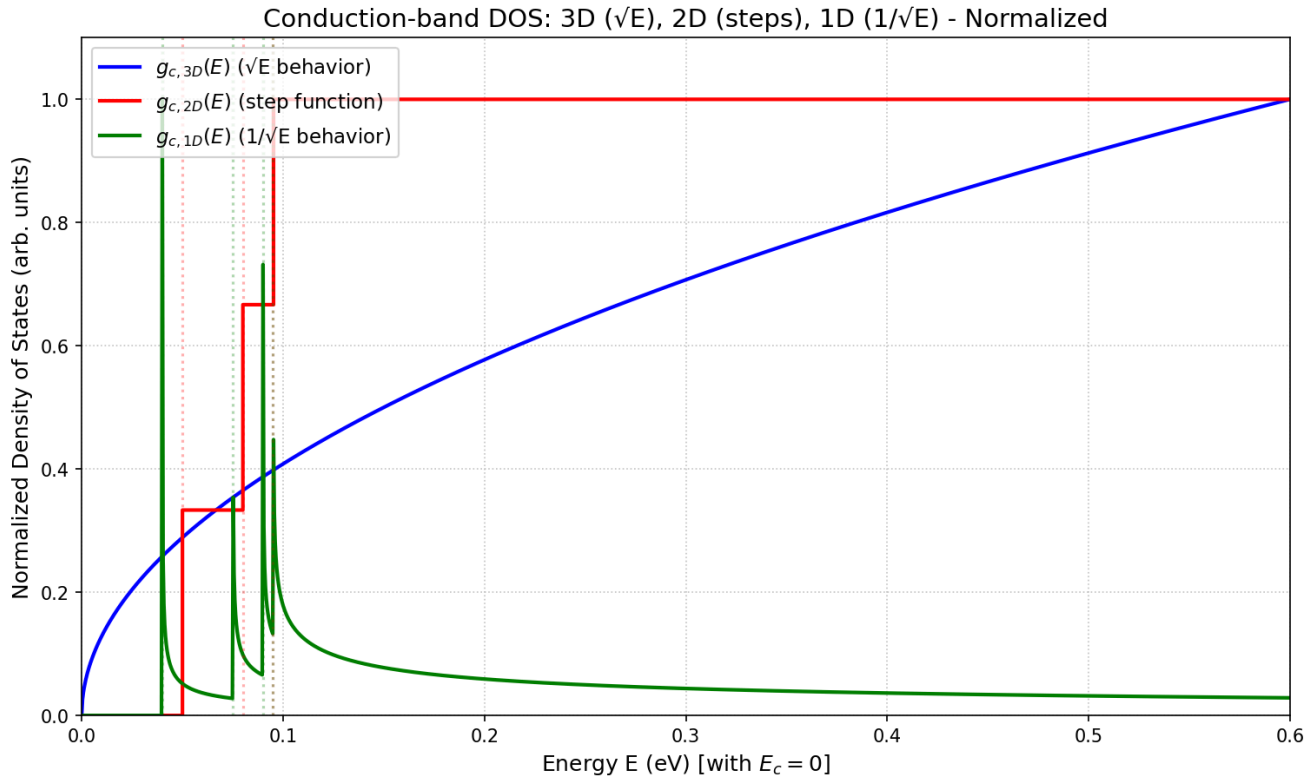
1D Density of States:

$$g_{c,1D}(E) = \sum_i \frac{2m^*}{\pi\hbar\sqrt{2m^*(E - E_c - \varepsilon_i)}}\theta(E - E_c - \varepsilon_i)$$

Units: 3D [states/(eV·cm³)], 2D [states/(eV·cm²)], 1D [states/(eV·cm)]

Visualization Note:

The plot shows normalized density of states to clearly visualize all three behaviors on the same scale.
The actual magnitudes differ significantly due to different dimensionalities and units.
Vertical dotted lines indicate subband energies for 2D (red) and 1D (green) systems.



Normalized density of states comparison showing characteristic behaviors: 3D (√E), 2D (step function), 1D (1/√E divergence). Vertical lines mark subband edges.

Problem 5: Electron and Hole Energy Densities (4.0 points)

Problem Statement:

Write a script (Python) to plot the number of electrons and holes per unit volume per unit energy as a function of energy.

Tasks:

(a) Plot electron AND hole energy densities vs. energy for E_F = E_C − 0.1 eV

(b) Plot electron AND hole energy densities vs. energy for E_F = E_V + 0.1 eV

(c) Compute electron densities for two Fermi levels via numerical integration and MB approximation

(d) Repeat (c) for holes

(e) Discuss when MB and Fermi–Dirac agree/differ

Electron Energy Density:

$$n(E) = g_c(E) \cdot f(E)$$

Hole Energy Density:

$$p(E) = g_v(E) \cdot [1 - f(E)]$$

Fermi-Dirac Distribution:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

Maxwell-Boltzmann Approximations:

$$n = N_C \exp\left(\frac{E_F - E_C}{k_B T}\right)$$

$$p = N_V \exp\left(\frac{E_V - E_F}{k_B T}\right)$$

Numerical Integration vs Maxwell-Boltzmann Comparison

Part (c): Electrons at 300 K (states/cm³)

EF = Ec - 0.10 eV: n_numerical = 6.671e+17, n_MB = 6.722e+17

EF = Ec - 0.01 eV: n_numerical = 1.796e+19, n_MB = 2.185e+19

Part (d): Holes at 300 K (states/cm³)

EF = Ev + 0.10 eV: p_numerical = 3.794e+17, p_MB = 3.823e+17

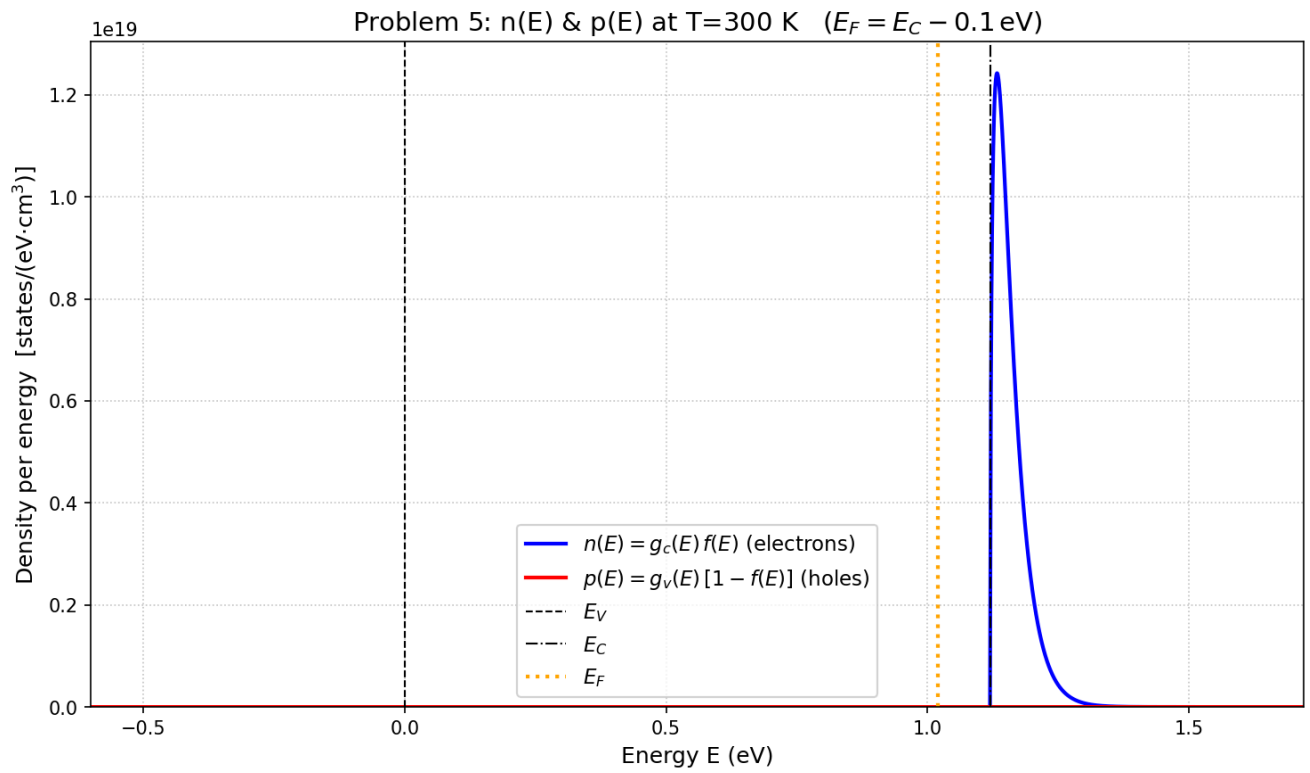
EF = Ev + 0.01 eV: p_numerical = 1.021e+19, p_MB = 1.243e+19

Fermi Level	Method	Electron Density (cm^-3)	Hole Density (cm^-3)
EF = Ec - 0.10 eV	Numerical	6.671e+17	-
EF = Ec - 0.10 eV	Maxwell-Boltzmann	6.722e+17	-
EF = Ec - 0.01 eV	Numerical	1.796e+19	-
EF = Ec - 0.01 eV	Maxwell-Boltzmann	2.185e+19	-
EF = Ev + 0.10 eV	Numerical	-	3.794e+17
EF = Ev + 0.10 eV	Maxwell-Boltzmann	-	3.823e+17
EF = Ev + 0.01 eV	Numerical	-	1.021e+19
EF = Ev + 0.01 eV	Maxwell-Boltzmann	-	1.243e+19

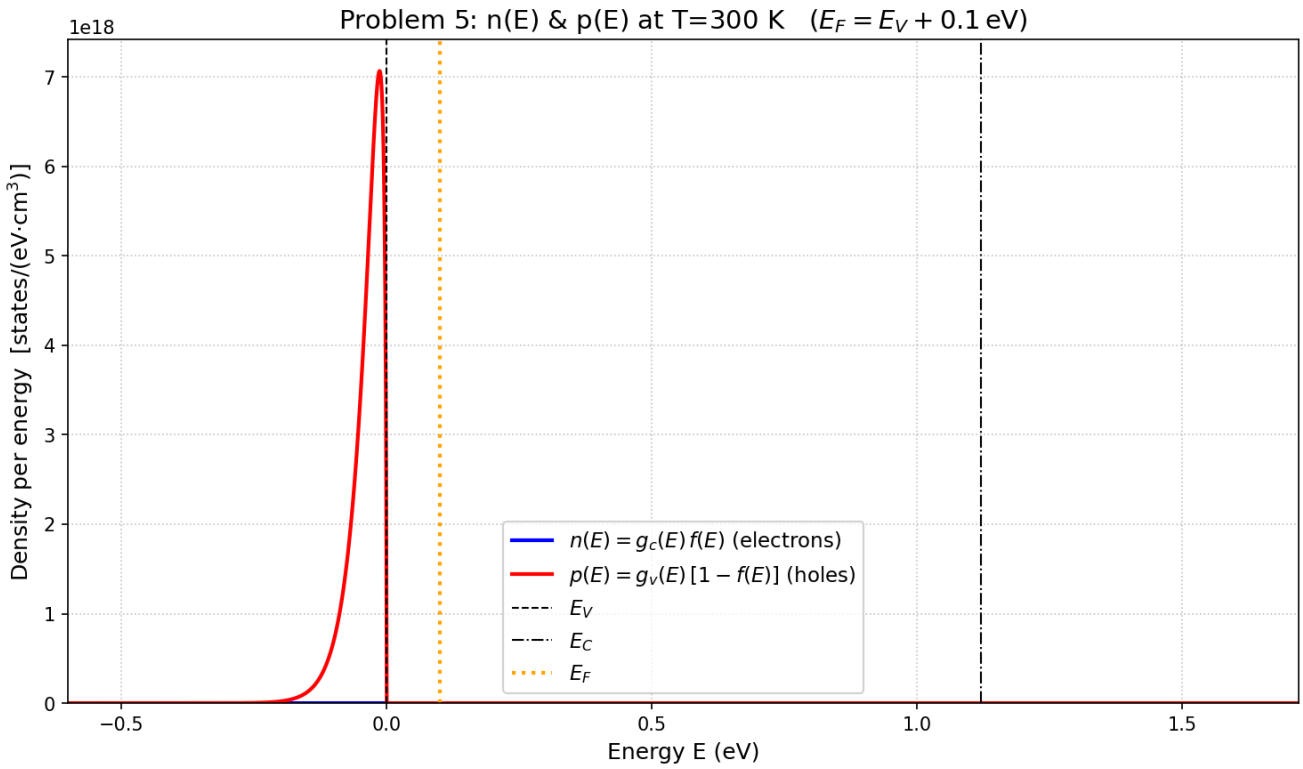
Maxwell-Boltzmann vs Fermi-Dirac Analysis

Part (e) Discussion:

MB vs Fermi–Dirac: When |E_F - band edge| ≳ ~3–4 k_BT (non-degenerate), occupation is small (f << 1 or 1-f << 1), so MB agrees with FD; as E_F approaches a band edge within ~k_BT, degeneracy grows and MB deviates.



Part (a): Energy density distributions showing $E_F = E_C - 0.1\,\text{eV}$



Part (b): Energy density distributions showing $E_F = E_V + 0.1\,\text{eV}$